



# Introduction to **Audio Content Analysis**

module 0.0: course introduction

alexander lerch

### ■ education

- Electrical Engineering (Technical University Berlin)
- Tonmeister (University of Arts Berlin)

### ■ professional

- Professor at the [Georgia Tech](#)
- previous: CEO at [zplane.development](#)

### ■ research focus

- Music Information Retrieval (MIR)
- Audio Content Analysis
- Audio Signal Processing
- Music Performance Analysis
- Music Generation



### Audio Content Analysis and Music Information Retrieval (MIR):

- extract and infer descriptors from music signals
- answers questions and tasks such as
  - “What is the tempo/key/mood of this song?”
  - “Transcribe this signal into a musical score.”
  - ...
- MIR is commercially interesting for, e.g.,
  - music recommendation
  - music identification
  - intelligent music production
  - automatic music generation

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after successful completion of this course, you will

- 1 have a good **overview of typical tasks** in MIR
- 2 **understand algorithmic approaches** in a large variety of basic MIR systems
- 3 be able to **implement MIR systems** in Matlab/Python
- 4 be able to **formally evaluate** systems with common datasets and metrics



### 1 Introduction to ACA and MIR

### 2 Fundamentals

- Signals & Pre-Processing
- Input Representations
- Inference
- Data & Evaluation

### 3 Music Transcription

- Tonal Analysis (Pitch, Key, ...)
- Analysis of Intensity
- Temporal Analysis (Onset, Beats, Structure, ...)
- Alignment

### 4 Music Identification & Classification

- Audio Fingerprinting
- Classification: Genre, Similarity, Mood, Instrument
- Music Performance Assessment



- basic knowledge in **DSP**
  - signals & systems, block diagrams, linear algebra, ...
- familiarity with **Matlab and Python**
  - scripting and functions, file I/O, ...
- helpful: knowledge of **machine learning** concepts
  - classification & regression, training and testing, evaluation metrics



# introduction

## course materials & resources

### ■ **text book:** “An Introduction to Audio Content Analysis”:

- 2nd edition
- 1st edition

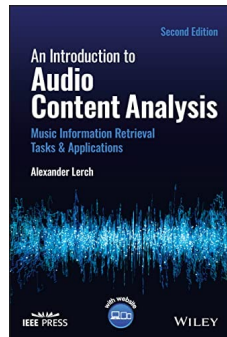
### ■ **optional reading**

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### ■ **online resources** @AudioContentAnalysis.org:

- slides & videos of previous classes
- datasets
- code (matlab, python, C++)

### ■ **needed software:** Python 3, (Matlab)





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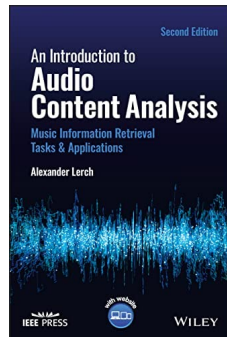
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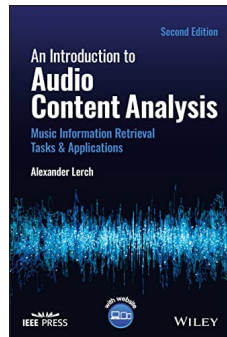
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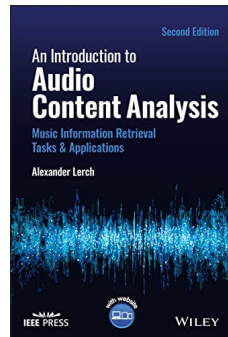
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### ■ grades

|                                       |     |
|---------------------------------------|-----|
| exercises & assignments               | 45% |
| exercises & participation             | 10% |
| project                               | 45% |
| presentation (proposal)               | 5%  |
| presentation (midterm)                | 5%  |
| presentation (final)                  | 5%  |
| paper                                 | 10% |
| algorithmic design and implementation | 20% |

### ■ **bonus points** for finding errors in the book or in the slides added to the assignment grade of your choice

- typos: 0.5 points
- language: 1 point
- misleading or incomplete statements: 2 points
- wrong statements and errors in equations: 3 points

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